



Information Architecture

What is Accessibility?

Accessibility is a broad term in UI/UX design. Making a design accessible means that all the members of the target audience group have equal access to the product or service. It also means that the design is usable and useful in multiple settings, meeting the varying needs of users. The product or services should be designed so that even users with disabilities can have the same user experience as others.

Accessibility means enabling people with diverse abilities and methods of access to effectively use your interface. The aim of accessibility is to remove barriers for perceiving, understanding and navigating your interface and ensure that nobody is excluded.

Accessibility is Essential for few but usable for all.

Accessibility Guiding Principles

There are four main guiding principles of accessibility upon which WCAG has been built. These four principles are known by the acronym POUR for **perceivable**, **operable**, **understandable** and **robust**. POUR is a way of approaching web accessibility by breaking it down into these four main aspects. Many of the technology challenges faced by disabled people/people with disabilities can be described using one of the POUR principles.

Perceivable

Means the user can identify content and interface elements by means of the senses. For many users, this means perceiving a system primarily visually, while for others, perceivability may be a matter of sound or touch.

Perceivable problem examples:

- A website's navigation consists of a number of links that are displayed in a different order from page to page. If a user has to relearn basic navigation for each page, how can user effectively move through the website?
- A Word document contains a number of non-English words and phrases. If the languages are not indicated, how can assistive technology present the text correctly?

- **Perceivable solutions:**

- **Text Alternatives**

Provide text alternatives for any non-text content so that it can be changed into other forms people need, such as large print, braille, speech, symbols or simpler language.

- **Time-based Media**

Provide alternatives for time-based media. (Anything that's embedded on a website that: moves, makes noise, displays content synchronized with another element of the site, or displays content that changes without user input.)

- **Adaptable**

Create content that can be presented in different ways (for example simpler layout) without losing information or structure.

- **Distinguishable**

Make it easier for users to see and hear content including separating foreground from background.

Operable

Means that a user can successfully use controls, buttons, navigation, and other interactive elements. For many users this means using assistive technology like voice recognition, keyboards, screen readers etc.

Operable problem examples:

- Mouse-dependent web content will be inaccessible to a person cannot use a standard mouse.
- People with low or no vision also rely on the functionality of the keyboard. They may be able to manipulate a mouse just fine, but it doesn't do them much good because they can't see where to click on the screen. The keyboard is much easier for a person who is blind to manipulate.

- **Operable solutions**
- **Keyboard Accessible**

Make all functionality available from a keyboard.

Keyboard accessibility is one of the most important principles of Web accessibility because it cuts across disability types and technologies

- **Enough Time**

Provide users enough time to read and use content.

- **Seizures**

Do not design content in a way that is known to cause seizures.

- **Navigable**

Provide ways to help users navigate, find content, and determine where they are.

Understandable

Users should be able to comprehend the content, and learn and remember how to use your OER (Open Educational Resources) site. Your OER should be consistent in its presentation and format, predictable in its design and usage patterns, and appropriate to the audience in its voice and tone.

Understandable problem examples:

- A website's navigation consists of a number of links that are displayed in a different order from page to page. If a user has to relearn basic navigation for each page, how can they effectively move through your OER?
- A site makes use of numerous abbreviations, acronyms, and jargon. If these are never defined, how can users with disabilities (and others) understand the content?

- **Understandable solutions**
- **Readable**

Make text content readable and understandable.

- **Predictable**

Make Web pages appear and operate in predictable ways.

- **Input Assistance**

Help users avoid and correct mistakes.

Robust

Content must be robust enough that it can be interpreted reliably by a wide variety of users, allowing them to choose the technology they use to interact with websites, online documents, multimedia, and other information formats. Users should be allowed to choose their own technologies to access OER content.

Robust problem examples:

- A website requires a specific version of a web browser to make use of its features. If a user doesn't or can't use that browser, how can that user experience the features of the site?
- A document format is inaccessible to a screen reader on a particular operating system. If a user employs that OS for day-to-day tasks, how can user gain access to the document?

Robust solutions:

- Maximize compatibility with current and future user agents, including assistive technologies.

Few points of accessibility to consider while designing...

Provide sufficient contrast between foreground and background

Foreground text needs to have sufficient contrast with background colors. This includes text on images, background gradients, buttons, and other elements. This does not apply for logos, or incidental text, such as text that happens to be in a photograph.

Example: Contrast ratio

✖ Insufficient

Some people cannot read text if there is not sufficient contrast between the text and background. For others, bright colors (high luminance) are not readable; they need low luminance.

✔ Sufficient

Some people cannot read text if there is not sufficient contrast between the text and background. For others, bright colors (high luminance) are not readable; they need low luminance.

Benefits: This helps people with low vision who often have difficulty reading text or perceiving graphics that have insufficient contrast.

Don't use color alone to convey information

While color can be useful to convey information, color should not be the only way information is conveyed. When using color to differentiate elements, also provide additional identification that does not rely on color perception. For example, use an asterisk in addition to color to indicate required form fields, and use labels to distinguish areas on graphs.

Example: Using color to convey meaning

✖ Color only

Required fields are in red

Name

Email

✔ Color and symbol

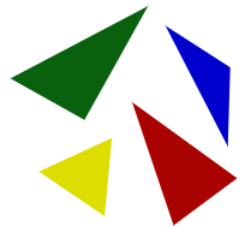
Required fields are in red and marked with an *

Name

Email *

Example: Refer to something using color alone

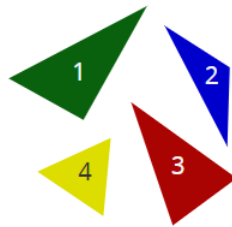
✖ Color only



Which is the right-angled triangle?

- ☐ Green
- ☐ Blue
- ☐ Red
- ☐ Yellow
- ☐ Don't know

✔ Color and number



Which is the right-angled triangle?

- ☐ Green (1)
- ☐ Blue (2)
- ☐ Red (3)
- ☐ Yellow (4)
- ☐ Don't know

Benefits: It is helpful for users with partial sight or limited color vision, color-blindness and for users who use text-only, limited color, or monochrome display screens.

Ensure that interactive elements are easy to identify

Provide distinct styles for interactive elements, such as links and buttons, to make them easy to identify. For example, change the appearance of links on mouse hover, keyboard focus, and touch-screen activation. Ensure that styles and naming for interactive elements are used consistently throughout the website.

Example: Unique styles for different link states

✔ Style links to stand out from text

Some people can't use a mouse and use only a [keyboard to navigate](#) through web pages.

It is important that users can reach all interactive elements using the keyboard, and that it is clear which element has focus.

Visible keyboard focus could be a border or highlight that moves as you tab through the web page.

✔ Mouse hover style

[keyboard to navigate](#)



✔ Keyboard focus style

[keyboard to navigate](#)

✔ Touch or click style

[keyboard to navigate](#)



Provide clear and consistent navigation options

Ensure that navigation across pages within a website has consistent naming, styling, and positioning. Provide more than one method of website navigation, such as a site search or a site map. Help users understand where they are in a website or page by providing orientation cues, such as breadcrumbs and clear headings.

[Cart](#) > [Billing](#) > [Shipping](#) > **Payment**



[Cart](#) / [Billing](#) / [Shipping](#) / **Payment**



Benefits: It is helpful for users with cognitive limitations, low vision and intellectual disabilities as it becomes easier to predict where they can find things on each page.

Ensure that form elements include clearly associated labels

Ensure that all fields have a descriptive label adjacent to the field. For left-to-right languages, labels are usually positioned to the left or above the field, except for checkboxes and radio buttons where they are usually to the right. Avoid having too much space between labels and fields.

✓ Example: Labels and input fields associated by proximity

Add a comment

Your E-mail

☐ I am happy for you to contact me

Your Website

Comment

Provide easily identifiable feedback

Provide feedback for interactions, such as confirming form submission, alerting the user when something goes wrong, or notifying the user of changes on the page. Instructions should be easy to identify. Important feedback that requires user action should be presented in a prominent style.

 **Example: Using error list, icon, and background color to make errors stand out**

Please correct the following errors:

1.  [Email address is invalid](#)
2.  [A Comment is required](#)

Add a comment

Required fields are in red and marked with an *

Name	Superbear
 E-mail *	superbear@@hq.example.com
Website	
 Comment *	

Use headings and spacing to group related content

Use whitespace and proximity to make relationships between content more apparent. Style headings to group content, reduce clutter, and make it easier to scan and understand.

Example: Spacing highlights relationship between content

✖ Little spacing and unclear relationship

Main heading

Horizontal lines representing text content.

Sub heading

Horizontal lines representing text content.

Sub heading

Horizontal lines representing text content.

✔ More spacing and clearer relationship

Main heading

Horizontal lines representing text content.

Sub heading

Horizontal lines representing text content.

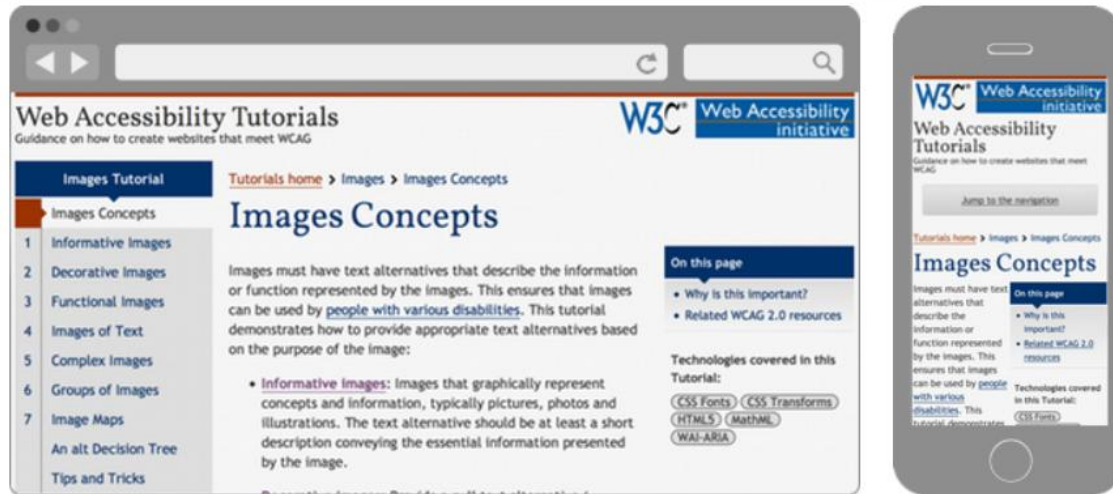
Sub heading

Horizontal lines representing text content.

Create designs for different viewport sizes

Consider how page information is presented in different sized viewports, such as mobile phones or zoomed browser windows. Position and presentation of main elements, such as header and navigation can be changed to make best use of the space. Ensure that text size and line width are set to maximize readability and legibility.

Example: Content and navigation adapt to smaller mobile screen



Display in a wide window with small text uses multiple columns for primary content, visible navigation options, and visible secondary information.

Display in a narrow window, such as a mobile phone, or with large text uses single column for primary content, navigation options are revealed using an icon, and secondary information is also revealed via icon.

Benefits: This allows the content to zoom and respond to various screen sizes without loss of information or functionality and without scrolling in two dimensions, consequently, helping the users with low vision, physical or motor disabilities.

- Remember, achieving complete accessibility is an ongoing process. While a website might be accessible today, changes or updates can affect this status. Regular testing and user feedback are crucial to maintaining an accessible web presence.



Thank you.